CS3000 Algorithms & Data **Practice Quiz 5D**	**Name:** _____ **NUID:** _____

Instructions:

- Please put your phones on silent and in your bag for the duration of the quiz.
- This quiz is designed to last 40 minutes.
- If you have a question, alert a proctor quietly, and we'll be glad to assist.
- If we can't read your answer, we can't grade your answer. Please write legibly.
- Practice version: harder than the real quiz; recognizing which class DP applies is the point.

Q1. A palindrome reads the same forwards and backwards (e.g. `level`). Given a string s of length n, we want the minimum number of characters that must be *inserted* (anywhere) to make s a palindrome.

(a) Explain why the answer is $n - \text{LCS}(s,\ s^R)$, where s^R is s reversed.

(b) For $s =$ `ABCAB` (so $s^R =$ `BACBA`), fill in the LCS table bottom-up.

$s_i \downarrow\ s_j^R \rightarrow$	0	1 (B)	2 (A)	3 (C)	4 (B)	5 (A)
0						
1 (A)						
2 (B)						
3 (C)						
4 (A)						
5 (B)						

(c) How many insertions are needed? Give one shortest palindrome you can build by inserting characters into `ABCAB`.

(d) What is the running time of this method in terms of n?

Q1 (continued).

Q2. You keep your own record of card transactions and compare it to the bank's statement at the end of the month. Both lists are in the same chronological order. A *discrepancy* is one of: (i) a transaction only in your record, (ii) a transaction only on the bank statement, or (iii) a transaction both sides have but with different amounts.

$$\text{Your record } R = (\$18, \$30, \$7, \$45, \$20), \qquad \text{Bank } B = (\$18, \$7, \$48, \$20, \$9).$$

(a) Model this as a DP you have seen in class. Define $D(i, j)$ in words and write its recurrence. What does each kind of discrepancy correspond to?

(b) Fill in the table bottom-up.

$R_i \downarrow \ B_j \rightarrow$	0	\$18	\$7	\$48	\$20	\$9
0						
1 (\$18)						
2 (\$30)						
3 (\$7)						
4 (\$45)						
5 (\$20)						

(c) What is the fewest number of discrepancies that explains the difference? List them.

Q2 (continued).